

Meeting Summaries of Jan 22

Date: January 22

Time: 14:30–15:05 PM(PDT)

Attendees: Dr. Javed, Dr. Yue and Feng

Topic: Discussion on Feng's reproduction work on Jian's thesis results.

Summary of Discussion

- Feng showed the side-by-side comparison of the results, tables and figures, that look similar to Jian's thesis results. Prof. Javed specifically **cross-checked** some results, which are:

1. **Table 1** - The Pollution PCs table.

Do the two tables have same shape and how the loadings of the same pollution variable look like across the two tables?

Do the PCs in the two tables represent similar pollution patterns?

2. **Table 2** - The ANOVA table of the taxa over the two clusters.

Are the significant taxa in Jian's thesis also significant in Feng's reproduction work?

What are the specific **SS Between** and **SS Within** of the F-test in Feng's results?

It is necessary to reorganize Feng's tables in the same format as Jian's tables if all numbers are computed in the same way.

3. **Table 3** - The ANOVA table of the environmental variables over the two clusters.

Are the (2) clusters in Feng's results also significantly different in environmental variables?

Are the differences over clusters in Feng's results similar to those in Jian's thesis?

It is better to reorganize Feng's tables in the same format as Jian's tables for easy comparison.

4. **Table 4** - The confusion matrix of the LDA trained on the chosen taxa clusters.

Are the two selected taxa clusters in Feng's results similar to those in Jian's thesis?

How do the classification accuracies compare between Feng's results and Jian's thesis?

If Feng thought the selection of clusters or other process is not very solid, Feng needs to re-produce the raw results first, then explain what the ideal should be and how that results look like.

5. **Figures** - The two regression of **ZCI** vs **Pollution Score** over the two clusters.

Feng needs to use same transformation method to coordinate the sites in the regression panel as Jian did.

The reproduction results should look similar at least in trend if not exactly the same.

Action Items

- Feng is going to align his parameters in computation and also the formats of tables and figures so that the results are computed and presented in the 'same' way as Jian's thesis.
- On top of the 'exact' reproduction, Feng is going to pinpoint where the extension he has made, so that the contributions can be clearly identified.

Follow-Up

Temporary meeting time: Jan 29, 14:30–15:00 PM(PDT)

Supportive materials

- A meeting recording with subtitles has been restored on Google Drive.